

Software Design Document

SweetLedger



SWEETLEDGER

Version 1.6 (Updated as of 12/05/25)

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TABLE OF CONTENTS

1. Introduction	
1.1. Purpose	3
1.2. Scope	3
1.3. Overview	3
1.4. Reference Material	3
1.5. Definitions and Acronyms	4
2. System Overview	4
3. System Architecture	
3.1. Architectural Design	5
3.2. Database Diagram	6
3.3. Decomposition Description	7
3.4. Design Rationale	7
4. Data Design	
4.1. Data Description	8
4.2. Data Dictionary	9
5. User Interface Design	
5.1. Overview of User Interface	14
5.2. Screen Images	16
5.3. Screen Objects and Actions	24
6. Requirements Matrix	31

1. INTRODUCTION

1.1 Purpose

This software design document describes the architecture and system design for SweetLedger, a web-based accounting application. SweetLedger is designed to help businesses manage financial transactions, generate reports, and maintain accurate accounting records. This document is intended for project managers, the development team, and other stakeholders involved in the implementation of the system.

1.2 Scope

This document details the implementation of the SweetLedger Web Application. The application is a secure, role-based system hosted on Cloudflare and uses MongoDB for data persistence. SweetLedger will consist of major functions including: Authentication/Security, Chart of Accounts Management, Transaction Journalizing, Transaction Posting/Review, Financial Reporting, and Ratio Analysis. The design will ensure the application is accessible from various devices and is fully functional across popular web browsers.

1.3 Overview

This document serves as the formal technical specification for the SweetLedger application. It is organized to lead the reader through the system's design. Section 2 (System Overview) outlines the major project requirements, user roles (Administrator, Manager, Accountant), and core accounting capabilities. Section 3 (System Architecture) defines the three-tier design. Section 4 (Data Design) specifies the document structure and constraints for accounting data within MongoDB. Section 5 (Component Design) details the functionality and interfaces of the C# backend and JavaScript frontend modules. Section 6 (Human Interface Design) covers the design and usability for the web interface across various devices. Finally, Section 7 (Requirements Traceability) confirms that all original project requirements are addressed by the design.

1.4 Reference Material

"Getting Started." *React.dev*, 17 Jul. 2019, react.dev/learn/getting-started.

MongoDB. "Getting Started." *MongoDB Documentation*, MongoDB, 1 Oct. 2025, www.mongodb.com/docs/get-started/. Accessed 1 Oct. 2025.

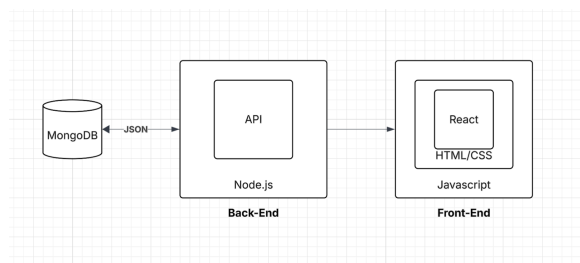
1.5 Definitions and Acronyms

Acronym	Meaning
API	Application Programming Interface
COA	Chart of Accounts
SDD	Software Design Document
DB	Database
PR	Post Reference
PK	Primary Key
FK	Foreign Key

2. SYSTEM OVERVIEW

SweetLedger is designed as a user-friendly web application for managing accounting tasks. Users authenticate via secure login to access functionality dependent on their role, such as viewing existing accounts, journalizing transactions, approving transactions, viewing financial reports, and performing ratio analysis. The frontend is built using React (JavaScript, HTML, CSS) for a responsive, modern, and interactive UI. The backend logic, workflows, and APIs are also implemented in JavaScript with Node.js. MongoDB stores structured data (users, accounts, transactions, logs). The system is deployed on Cloudflare at sweetledger.com using Cloudflare for our frontend and Render for our backend. Built-in help and a tab-driven interface assist users in navigation. Administrators, managers, and regular users see different views and permissions. *Figure 1* shows the architectural structure we've chosen for the development of SweetLedger

Figure 1



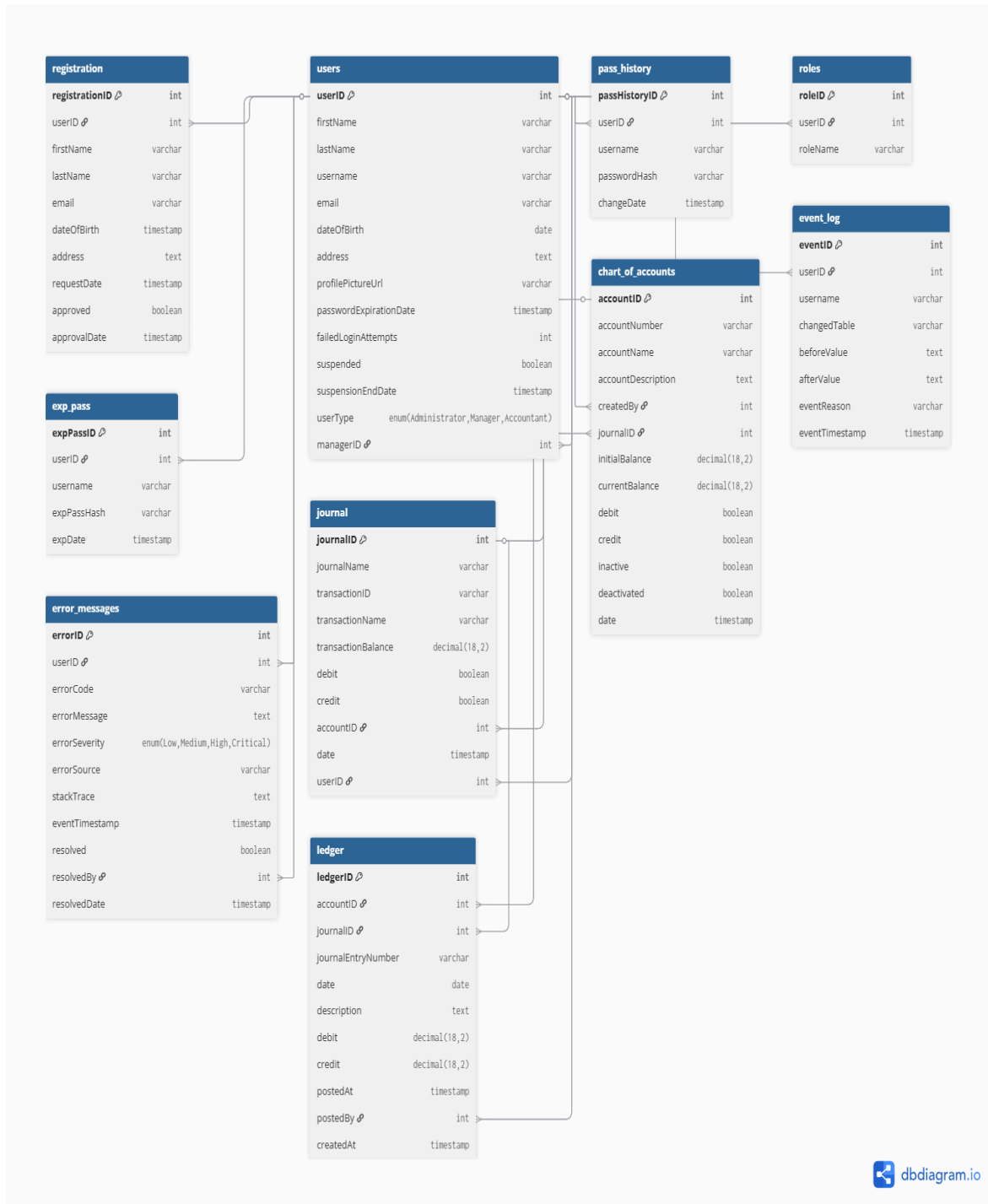
3. SYSTEM ARCHITECTURE

3.1 Architectural Design

To support the system's functionality, responsibilities were divided into high-level subsystems that interact seamlessly to provide a secure and user-friendly interface. The User class represents individuals who interact with the system and contains attributes such as personal information, login credentials, and account status. Each user is associated with a Role, which determines their permissions and accessible features (Administrator, Manager, or Regular User). The PasswordPolicy class enforces authentication rules, ensuring that all credentials meet security standards before being accepted into the system. The Account Management/Account View module allows users to view current users and accounts. The Chart of Accounts and Journalizing & Ledger Module enable users (based on role permissions) to manage, create, and approve financial entries, and maintain accurate ledgers. The Event Logs module shows pre- and post-conditions of any update made in the system, such as when a new user is registered. The Financial Reports Module enables users, dependent on their role, to generate reports such as the Trial Balance or Balance Sheets.

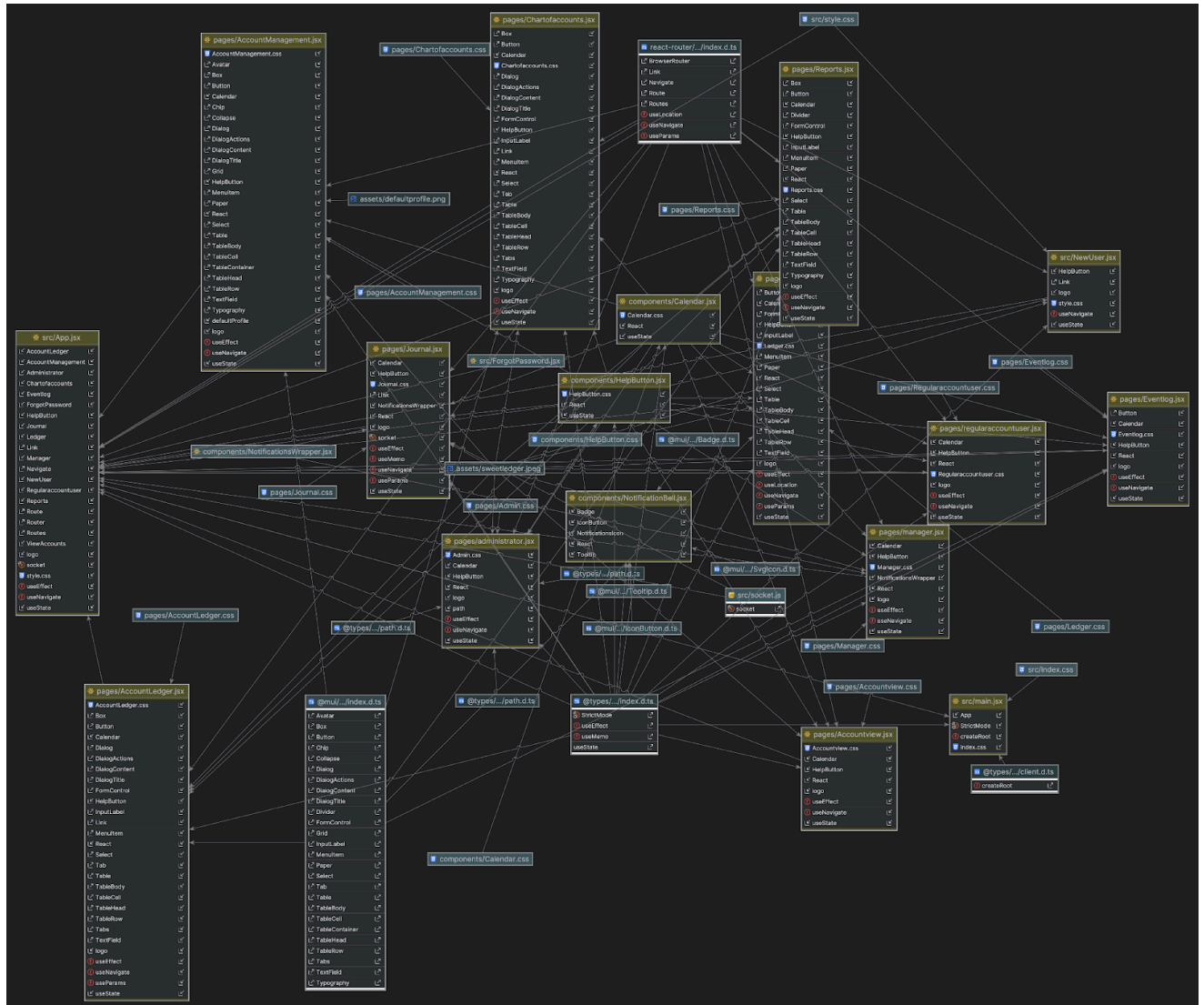
3.2 Database Diagram

Figure 2: Database Diagram



3.3 Decomposition Description

Figure 3: UML Class diagram showing the relationship between the different subsystems.



3.4 Design Rationale

The chosen three-tier architecture (User Interface, API, and Data layers) was selected for its scalability, maintainability, and separation of concerns. Using React for the UI ensures responsiveness across devices, while Node.js APIs hosted on Render provide reliable backend logic and integration. MongoDB was chosen for its flexibility in handling dynamic user and document data.

4. DATA DESIGN

4.1 Data Description

4.1.1 User data is found in the **users** collection, which stores each system user's username, hashed password, first name, last name, email, time created, account status and more. This allows the system to maintain login credentials securely, enforce password policies, and track account status, such as active, inactive, or suspended. When a new user is created, the **registration** collection handles their input through the NewUser Component and the form data created when the user enters their information.

4.1.2 The **chart_of_accounts** table stores all account records. Each account record includes details such as account name, account number, description, normal side, category, subcategory, initial balance, debit, credit, balance, date and time added, user ID of the creator or modifier, comment, and status. The system enforces data integrity by ensuring that duplicate account names or numbers are not allowed, and account numbers contain only numeric values. Additionally, accounts with non-zero balances cannot be deactivated, maintaining the integrity of financial data.

4.1.3 The **journal** table captures all journal entries created by managers or accountants. Each entry includes a journal entry ID, associated account IDs, description, date, debit and credit amounts (supporting multiple debits and credits per entry), attached source documents (PDF, Word, Excel, CSV, JPG, PNG), status (pending, approved, or rejected), and the user IDs of both the creator and approver with timestamps. The system validates that total debits equal total credits and that each entry contains at least one debit and one credit, preventing submission of invalid transactions. The **ledger** table records all postings from approved journal entries to account ledgers. Ledger entries include ledger entry ID, account ID, journal entry ID (post reference), date, description, debit, credit, balance, user ID of the posting initiator, and timestamp. Ledger balances are automatically updated after each approved journal entry, and users can filter and search ledger entries by account, date, or amount to ensure accuracy and ease of access.

4.1.4 To ensure accountability and traceability of all data changes, the system maintains an **event_logs** collection that records every modification made to user or account data. Each event log entry contains a unique event ID, the type of entity affected, the ID of the modified record, the type of action performed (such as add, edit, or deactivate), the before and after snapshots of the record, the user ID of the individual who performed the action, and the timestamp of when the change occurred.

4.1.5 Error messages that are displayed when entries are invalid, user credentials are incorrect, or any system fault is found in the **error_messages** table.

4.2 Data Dictionary

Registration Table

Field Name	Data Type	Description
registrationID	int	PK: Unique ID number for that user's registration
userID	int	Unique ID number for user
firstName	varchar	User first name
lastName	varchar	User's last name
email	varchar	User's email
dateOfBirth	timestamp	User's date of birth
address	text	User's home address
requestDate	timeStamp	Date that user requested access to system/create new profile
approved	boolean	Indicates whether user is approved by admin to access system or not
approvalDate	timestamp	Date user was approved to access system

Exp_pass Table

Field Name	Data Type	Description
expPassID	int	PK: Unique ID of expired password
userID	int	FK: User ID associated with expired password
username	varchar	Username of user associated with expired password

expPassHash	varchar	Expired password in hash
expDate	timestamp	Date password expired

Error_messages Table

Field Name	Data Type	Description
errorID	int	PK: Unique ID number for error
userID	int	FK: ID number of user causing error
errorCode	varchar	Code of error
errorMessage	text	Error Message to be displayed
errorServerity	enum(Low,Medium, High, Critical)	How severe the error caused is to the system
errorSource	varchar	Where the error came from
stackTrace	text	The stack trace of the error
eventTimestamp	timestamp	When the error occurred
resolved	boolean	Indicates whether the error is resolved or not
resolvedBy	int	ID number of user who resolved error
resolvedDate	timestamp	Date error was resolved

Users Table

Field Name	Data Type	Description
userID	int	PK: Unique ID number associated with user
firstName	varchar	User first name
lastName	varchar	User last name

username	varchar	User's username
email	varchar	User's email address
dateOfBirth	date	User's date of birth
address	text	User's home address
profilePictureUrl	varchar	URL holding user's profile image
passwordExpirationDate	timestamp	Date password expires
failedLoginAttempts	int	Number of failed login attempts at the time of the session
suspended	boolean	Indicates whether user is suspended from accessing system or not
suspensionEndDate	timestamp	Date in which user is suspended from system
role	varchar	Role of user (Admin, Manager, Accountant)

Journal Table

Field Name	Data Type	Description
journalID	int	PK: Unique ID number of submitted journal entry
description	varchar	Description entered for the associated entry
status	varchar	Entry status (Pending, Rejected, Approved)
createdBy	varchar	Username of user who submitted entry
entries	Array	Array of object that contain accounts used in transaction

createdAt	timestamp	Time entry was submitted
reviewedBy	varchar	Username of manager who reviewed entry
reviewedAt	timestamp	Timestamp entry was reviewed at
credit	decimal(8,2)	Amount account was credited
debit	decimal(8,2)	Amount account was debited
accountId	int	Number of account used in transaction
accountName	varchar	Name of account used in transaction
date	date	Date of entry

Ledger Table

Field Name	Data Type	Description
ledgerID	int	PK: Unique ID number for transaction
accountID	int	FK: Account number associated with transaction
journalID	int	FK: Journal ID number for entry where transaction/ledger was derived from
date	date	Date of transaction
description	text	Description associated with the transaction
debit	decimal(8,2)	Amount account was debited
credit	decimal(8,2)	Amount account was credited
postedAt	timestamp	Time transaction was approved
postedBy	varchar	Username of user who

		approved transaction
createdAt	timestamp	Time in which ledger was created

Pass_History Table

Field Name	Data Type	Description
passHistoryID	int	PK: Unique ID for password history
userID	int	FK: User associated with password history
username	varchar	Username of user
passwordHash	varchar	Past password
changeDate	timestamp	Date password last changed

Chart_of_accounts Table

Field Name	Data Type	Description
accountID	int	PK: Unique ID number for each account
accountNumber	int	Account number
accountName	varchar	Name of account
accountDescription	text	Description/Comments for account
createdBy	varchar	Username of user who created account
initialBalance	decimal(8,2)	Initial balance of account at creation
currentBalance	decimal(8,2)	Current account balance
debit	decimal(8,2)	Amount debited to account

credit	decimal(8,2)	Amount credited to account
normalSide	varchar	Normal side of account (L/R)
deactivated	boolean	Indicated whether account is deactivated or not
category	varchar	Category/Type of account (eg. Asset, Liability..)
subcategory	varchar	Subcategory of account (eg. Current, Long-Term..)

Event_log Table

Field Name	Data Type	Description
eventID	int	PK: Unique ID number for each logged event
userID	int	FK: ID number of user who caused event
username	varchar	Username of user who caused event
changedTable	varhcar	Name of table that was altered
beforeValue	text	Value before field was altered
afterValue	text	Value after field was altered
eventReason	varchar	Reason for event
eventTimestamp	timestamp	Time of event occurrence

5. User INTERFACE DESIGN

5.1 Overview of User Interface

Upon navigating to the login page, users are presented with clearly labeled text boxes for entering

their username and password, along with buttons to submit their credentials, request a new account, or reset a forgotten password. Once successfully logged in, the user's name and profile picture appear prominently at the top-right corner of the screen, confirming their identity and role. The interface maintains a consistent layout across all modules, featuring a system logo and navigation bar at the top of each page, providing quick access to core services such as the Dashboard, Account Management, the Chart of Accounts, Journalize, Event Logs, and the Financial Reports Module. The Event Logs page displays the pre- and post-conditions for any changes in the system.

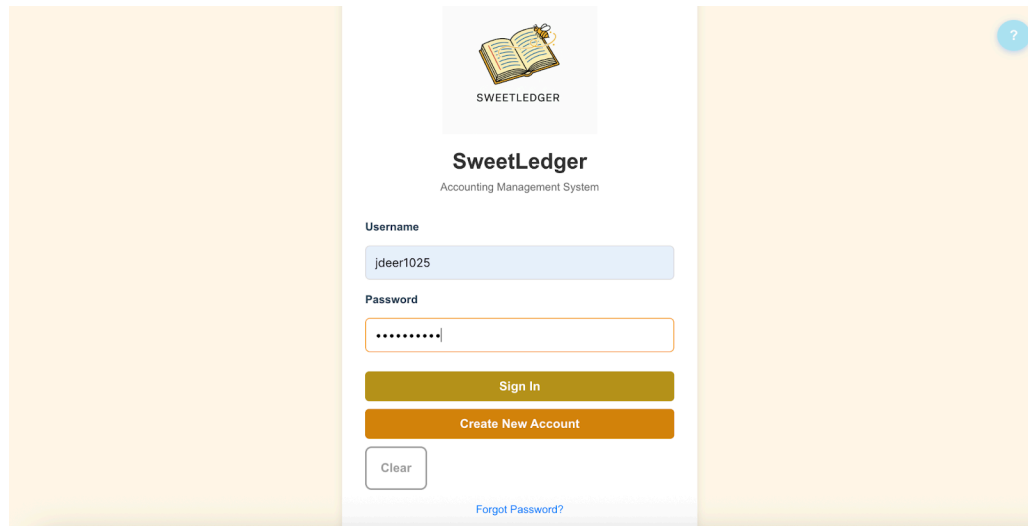
Administrators can create new users, update user information, activate or deactivate accounts, suspend users, view all system users, and send email notifications from the Account Management page. Within the Chart of Accounts component, administrators can add, edit, view, or deactivate accounts, with each account page containing structured fields for account name, number, category, subcategory, created by, created at, comments, and balance, along with automatic monetary formatting to two decimal places and comma separators. Administrators can also search for accounts by name or number, filter data by attributes such as category or balance, and generate reports summarizing account details.

Managers and accounting users interact with a slightly different view from the administrator. For users and accounts, both roles can view but not edit. In the chart of accounts, selecting an account number opens its corresponding Ledger Page, which displays past transactions that were made with the corresponding account. Both roles can access financial reports, filter or search entries, and navigate between the chart of accounts, ledgers, and journal entries. From the Journalizing Page, accountants can create and submit journal entries using accounts from the chart of accounts, attach source documents (PDF, Word, Excel, CSV, JPG, or PNG), and ensure that debits equal credits before submission. Once submitted, managers receive an automated notification and can approve or reject journal entries, entering comments for rejected transactions. Managers &

Accountants can filter entries by status type (pending, approved, rejected) using the tabs found at the top of the page and filter according to account name, number, balance, or submission date. Approved entries are automatically posted to the ledger, updating account balances in real time. Both managers and accountants can click the Post Reference (PR) within a ledger to view the journal entry that created it, ensuring complete traceability. The Financial Reports Page lets Managers & Accountants generate, print, save, or email reports such as the Balance Sheet, Income Statement, Statement of Retained Earnings, and the Trial Balance from a certain date.

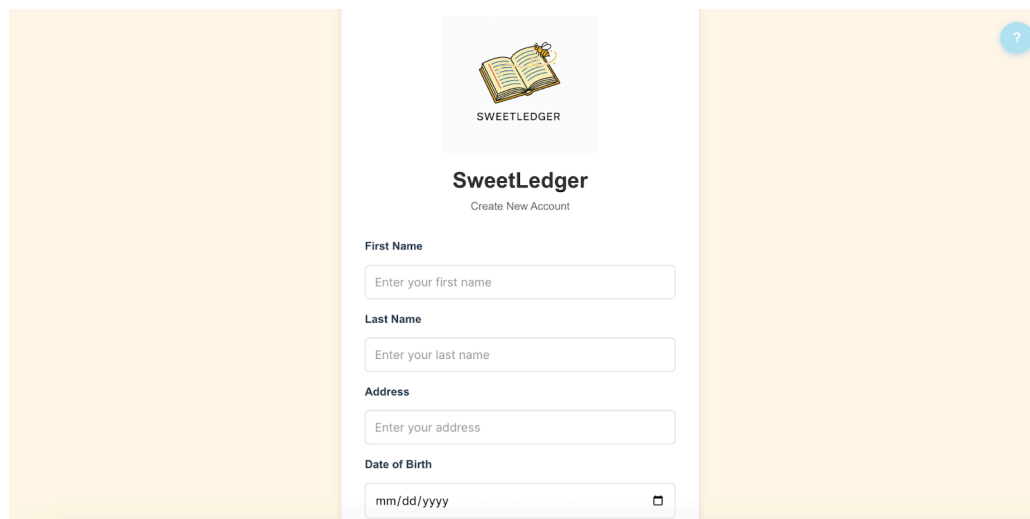
5.2 Screen Images

Figure 4: Login Page



The login page features a central white card on a light orange background. At the top of the card is the SweetLedger logo, which includes an icon of an open book with a key and the text 'SWEETLEDGER'. Below the logo is the 'SweetLedger' title and the subtitle 'Accounting Management System'. The form contains fields for 'Username' (with the value 'jdeer1025') and 'Password' (masked with dots). There are three buttons: 'Sign In' (orange), 'Create New Account' (orange), and 'Clear' (white with a grey border). A link for 'Forgot Password?' is located at the bottom of the card. A blue circular help icon with a question mark is in the top right corner of the page.

Figure 5: Create Account Page



The create account page has a similar layout to the login page. It features the SweetLedger logo and the title 'SweetLedger' with the subtitle 'Create New Account'. The form includes fields for 'First Name' (placeholder: 'Enter your first name'), 'Last Name' (placeholder: 'Enter your last name'), 'Address' (placeholder: 'Enter your address'), and 'Date of Birth' (placeholder: 'mm/dd/yyyy' with a calendar icon). A blue circular help icon with a question mark is in the top right corner of the page.

Figure 6: Reset Password Page



SWEETLEDGER

SweetLedger

Reset Your Password - Step 1 of 3

Username

Email

Next

Clear

[Back to Login](#)

?

Figure 7: Manager Dashboard Page

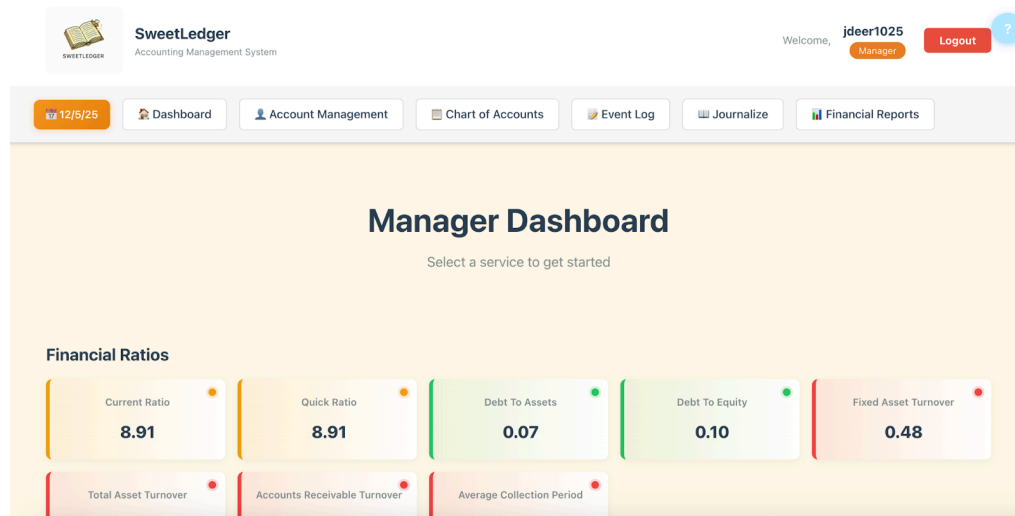


Figure 8: Accountant Dashboard Page

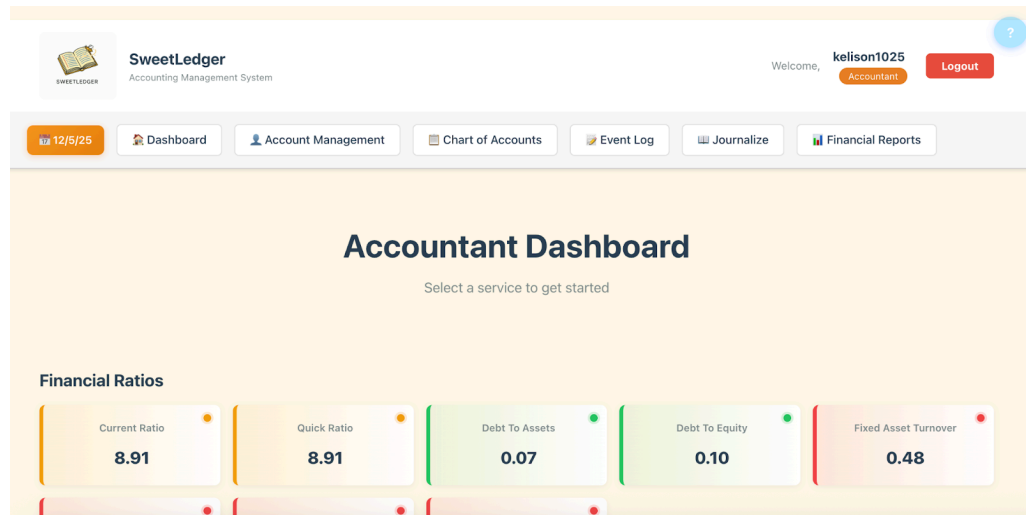


Figure 9: Administrator Dashboard Page

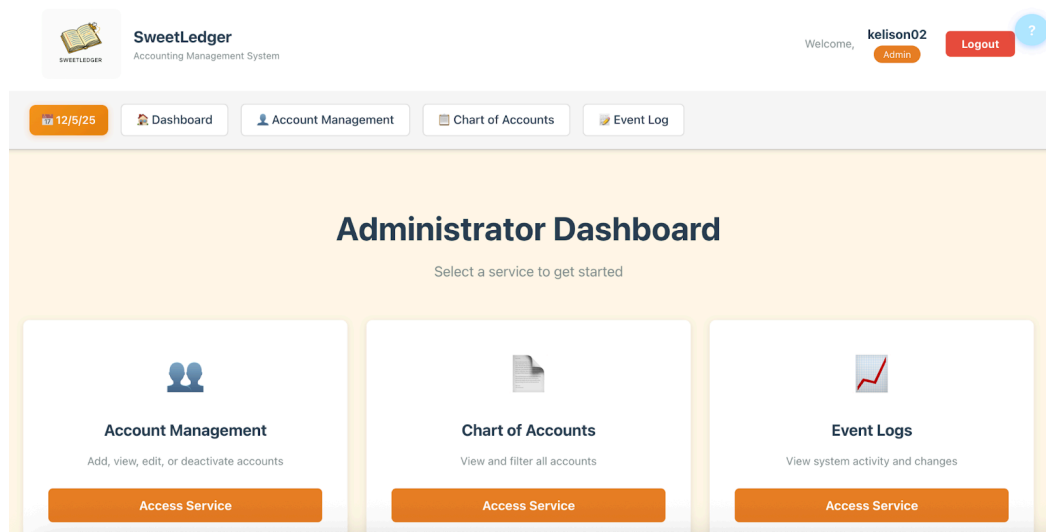


Figure 10: Account Management Page (Admin View)

Administrator Account Management

Welcome, **kelison02**
AdminLogout

12/5/25DashboardAccount ManagementChart of AccountsEvent Log

User ManagementView All Users ReportExpired Passwords ReportCreate New User

Username	Email	Role	Status	Actions
kelison02	kelison@students.kennesaw.edu	Admin	Active	EditDeactivateSuspendEmail
bgreen025	test@gmail.com	Accountant	Active	EditDeactivateSuspendEmail
jdeer1025	test@gmail.com	Manager	Active	EditDeactivateSuspendEmail
mcrowleyadmin1	matthewcrowley2002@gmail.com	Admin	Active	EditDeactivateSuspendEmail

Figure 11: Chart of Accounts Page (Admin View)

Chart of Accounts

Welcome, **kelison02**
AdminLogout?

12/5/25DashboardAccount ManagementChart of AccountsEvent Log

Search and Filter Accounts

Filter By
All Fields

Search

Search

Select a filter type and enter a search term to find accounts.

Chart of Accounts

Accounts ReportAdd Account

Manage your accounts here.

Select to Edit	Account Number	Account Name	Account Type	Subcategory	Balance	Created By	Date Created	Comments	
☒	1001	Cash	Asset	Current	8875	kelison02	2025-10-20T		SAVE CANCEL
☐	1012	Accounts Receivable	Asset	Current	\$3,450.00	kelison02	2025-10-20T00:00:00Z		
☐	1403	Supplies	Asset	Current	\$1,020.00	kelison02	2025-10-20T00:00:00Z		
☐	1414	Office Equipment	Asset	Long-Term	\$9,300.00	kelison02	2025-10-20T00:00:00Z		

Figure 12: Event Log Page


Event Log

Welcome, **kelison02**
Admin
Logout

12/5/25

Dashboard

Account Management

Chart of Accounts

Event Log

Event Log

View all account changes, including before and after states.

ID	User ID	Action	Timestamp ▼	Before	After
69335db463c73670446f9f3d	ttest1225	New user registered	12/5/2025, 10:33:24 PM		First Name: Test Last Name: Test Address: 123 test street DOB: 2002-12-06 Email: matthewcrowley2002@gmail.com Username: ttest1225 ID: 69335db463c73670446f9f3c

Figure 13: Account Management Page (Manager/Accountant View)



Account View

Welcome, **jdeer1025**
Manager Logout

12/5/25

Dashboard

Account Management

Chart of Accounts

Event Log

Journalize

Financial Reports

User Management

Username	Email	Role	Status
kelison02	kelison@students.kennesaw.edu	Admin	Active
bgreen025	test@gmail.com	Accountant	Active
jdeer1025	test@gmail.com	Manager	Active
mcrowleyadmin1	matthewcrowley2002@gmail.com	Admin	Active
mcrowleymanager1	matthewcrowley2002@gmail.com	Manager	Active

Figure 14: Chart of Accounts Page (Manager/Accountant View)



Chart of Accounts

Welcome, **jdeer1025**
Manager Logout

12/5/25

Dashboard

Account Management

Chart of Accounts

Event Log

Journalize

Financial Reports

Search and Filter Accounts

Filter By
All Fields

Search

Search

Select a filter type and enter a search term to find accounts.

Chart of Accounts

Accounts Report

Manage your accounts here.

Account Number	Account Name	Account Type	Subcategory	Balance	Created By	Date Created	Comments
1001	Cash	Asset	Current	\$8,875.00	kelison02	2025-10-20T00:00:00Z	
1012	Accounts Receivable	Asset	Current	\$3,450.00	kelison02	2025-10-20T00:00:00Z	
1403	Supplies	Asset	Current	\$1,020.00	kelison02	2025-10-20T00:00:00Z	
1414	Office Equipment	Asset	Long-Term	\$9,300.00	kelison02	2025-10-20T00:00:00Z	
1505	Prepaid Rent	Asset	Current	\$3,000.00	kelison02	2025-10-20T00:00:00Z	
1506	Prepaid Insurance	Asset	Current	\$1,650.00	kelison02	2025-10-20T00:00:00Z	
1807	Accumulated Depreciation	Asset	Long-Term	\$500.00	kelison02	2025-10-20T00:00:00Z	
2101	Accounts Payable	Liability	Current	\$1,000.00	kelison02	2025-10-20T00:00:00Z	
2203	Salaries Payable	Liability	Current	\$20.00	kelison02	2025-10-20T00:00:00Z	
2452	Unearned Revenue	Liability	Current	\$1,000.00	kelison02	2025-10-20T00:00:00Z	
3001	Service Revenue	Revenue		\$13,425.00	kelison02	2025-10-20T00:00:00Z	
3010	Contributed Capital	Equity		\$20,250.00	kelison02	2025-10-20T00:00:00Z	
5003	Advertising Expense	Expense		\$120.00	kelison02	2025-10-20T00:00:00Z	

Figure 15: Ledger Page


Ledger

Welcome, **jdeer1025** Manager Logout

12/5/25
Dashboard
Account Management
Chart of Accounts
Event Log
Journalize
Financial Reports

1001 - Cash

Start Date
mm/dd/yyyy

End Date
mm/dd/yyyy

CLEAR

DATE	POST REFERENCE	DESCRIPTION	DEBIT	CREDIT	BALANCE
2025-04-04	JE-1	Assets received by capital contributor	\$10,000.00		\$10,000.00
2025-04-04	JE-2	Paid three months rent on lease rental contract		\$4,500.00	\$5,500.00
2025-04-04	JE-3	Paid premium on property and casualty insurance policies for the year		\$1,800.00	\$3,700.00
2025-04-06	JE-4	Received cash from clients as an advanced payment for services to be provided	\$3,000.00		\$6,700.00
2025-04-07	JE-5	Received cash from clients on account	\$800.00		\$7,500.00
2025-04-11	JE-6	Paid cash for newspaper advertisement		\$120.00	\$7,380.00

Figure 16: Journalize Page

Journal Entries

Welcome, **jdeer1025**
ManagerLogout?

12/5/25DashboardAccount ManagementChart of AccountsEvent LogJournalizeFinancial Reports

+ New Entry

All (39)Pending (1)Approved (27)Rejected (11)

Entry Type: All Entries

Search by account name, amount, date, or description...

From: mm/dd/yyyy

To: mm/dd/yyyy

JE-e49cdb

Approved

Adjusting Entry

4/30/2025

Paying for Prepaid Insurance

Figure 17: Financial Reports Page

Financial Reports

Welcome, **jdeer1025**
ManagerLogout?

12/5/25DashboardAccount ManagementChart of AccountsEvent LogJournalizeFinancial Reports

Report Configuration

Report Type

Trial Balance

Report Date

12/06/2025

GENERATE REPORT

PRINTEMAILSAVE

SweetLedger Inc.

Trial Balance

As of 12/6/2025

ACCOUNT NUMBER	ACCOUNT NAME	DEBIT	CREDIT
1001	Cash	\$8,875.00	-
1012	Accounts Receivable	\$3,450.00	-
1403	Supplies	\$1,020.00	-
1414	Office Equipment	\$9,300.00	-
1505	Prepaid Rent	\$3,000.00	-
1506	Prepaid Insurance	\$1,650.00	-
1807	Accumulated Depreciation	-	\$500.00
2101	Accounts Payable	-	\$1,000.00
2203	Salaries Payable	-	\$20.00
2452	Unearned Revenue	-	\$1,000.00
3001	Service Revenue	-	\$13,425.00
3010	Contributed Capital	-	\$20,250.00
5003	Advertising Expense	\$120.00	-
5004	Salaries Expense	\$5,320.00	-
5005	Telephone Expense	\$130.00	-

5006	Utilities Expense	\$200.00	-
5007	Insurance Expense	\$150.00	-
5008	Supplies Expense	\$980.00	-
5009	Depreciation Expense	\$500.00	-
5010	Rent Expense	\$1,500.00	-
Total		\$36,195.00	\$36,195.00

5.3 Screen Objects and Actions

5.3.1 Login Page

- **Description:** This page authenticates users before granting access to the systems
- **Screen Objects:**
 - Username text box
 - Password text box
 - Submit button (“Login”)
 - “Forgot Password” button
 - “Create New User” button
 - Company logo (consistent branding across all pages)
- **User Actions:**
 1. Enter username and password.
 2. Click “Login” to submit credentials.
 3. Optionally select “Forgot Password” or “Create New User” for alternate workflows.
- **System Responses:**
 - Valid credentials: User is redirected to the Dashboard.
 - Invalid credentials: Error message displayed below input fields.
 - “Forgot Password” opens the password recovery page.
 - “Create New User” opens the new user registration page.

5.3.2 Create New User Page

- **Description:** This page enables new users to request access to the system by submitting required registration details.
- **Screen Objects:**
 - Input fields:
 - First Name
 - Last Name
 - Date of Birth
 - Address
 - Email
 - Desired Password
 - Auto-populated Username field
 - Submit button
- **User Actions:**
 1. Complete all input fields.
 2. Click “Submit” to send the new user request.
- **System Responses:**
 - Valid and complete form: System sends data to the backend for administrative approval and displays a confirmation message.
 - Missing or invalid data: Input error messages are shown next to the affected fields.

5.3.3 Forgot Password Page

- **Description:** This page allows users to securely reset their login credentials.
- **Screen Objects:**
 - Username input field
 - Email input field
 - Security question prompt
 - New password field (appears after validation)
 - Submit button
- **User Actions:**

1. Enter username and associated email.
2. Answer the displayed security question.
3. Enter and confirm a new password.
4. Click “Submit.”

- **System Responses:**

- If all information is valid, the password is updated, and a confirmation message is displayed.
- If validation fails, the user receives a descriptive error message.

5.3.4 Account Management Page (Administrator View)

- **Description:** This page provides administrators with tools to manage all user accounts in the system.

- **Screen Objects:**

- Dynamic data table listing all users (columns: Username, Email, Role, Account Status)
- Buttons: Update, Activate/Deactivate, Suspend, Send Notification
- “Create New User” button

- **User Actions:**

1. View user list and select a record for action.
2. Use action buttons to modify account status or update details.
3. Create a new user directly through the provided button.

- **System Responses:**

- System applies the selected action (e.g., suspend, activate) and confirms with a message.
- Updates are immediately reflected in the user table.

5.3.5 Account View Page (Manager and Accountant View)

- **Description:** This page provides a read-only version of the Account Management interface for Managers and Accountants.

- **Screen Objects:**

- Data table listing user information identical to the admin version (without action buttons).
- **User Actions:**
 1. View and search existing user accounts.
- **System Responses:**
 - Display of user records without editing capability.

5.3.6 Chart of Accounts Page

- **Description:** This page manages and displays all financial accounts in the system.
- **Screen Objects:**
 - Search bar
 - Filter panel (Category, Subcategory, Balance, Name, Number)
 - Data table (Account Name, Number, Category, Subcategory, Balance, Created By, Date Created, Comments)
 - “Select to Edit” column (Administrators only)
- **User Actions:**
 1. Search or filter accounts.
 2. View account details.
 3. (Administrators) Edit accounts using the “Select to Edit” column.
- **System Responses:**
 - Displays filtered or searched account data.
 - Automatically formats all monetary fields with commas and two decimal places.
 - Restricts edit access to authorized users.

5.3.7 Ledger Page

- **Description:** Displays detailed transaction history for a specific account.

- **Screen Objects:**
 - Data table (Date, Description, Debit, Credit, Post Reference (PR), Running Balance)
 - Filter controls (Date Range, Account Name, Amount)
 - Search bar
- **User Actions:**
 1. Click an account number from the Chart of Accounts to open its ledger.
 2. Search or filter transactions.
 3. Click a PR number to open the related Journal Entry.
- **System Responses:**
 - Loads all transactions for the selected account.
 - Navigates to the Journal Entry Page when a PR link is selected.

5.3.8 Journalize Page

- **Description:** Allows Managers and Accountants to create, review, and process journal entries.
- **Screen Objects:**
 - Journal Entry Table (dynamic debit and credit rows)
 - Description field
 - Account selection dropdown (populated from Chart of Accounts)
 - File upload control (PDF, Word, Excel, CSV, JPG, PNG)
 - Submit button
 - (Manager view) Approve button, Reject button, Comment field
 - Tab Panel (Pending, Approved, Rejected)
- **User Actions:**
 1. Enter debit and credit entries with descriptions.
 2. Attach supporting documentation.
 3. Submit journal entry for manager approval.
 4. (Manager) Review and approve or reject entries.
- **System Responses:**

- Valid entries: Sent for approval or posted to the ledger upon approval.
- Errors: Highlighted in red below affected fields.
- Approved entries: Automatically update related ledgers.
- Rejected entries: Labeled “Rejected” with visible indicators.

5.3.9 Event Logs Page

- **Description:** Displays a chronological record of system events for audit and traceability.
- **Screen Objects:**
 - Data table with columns: Event ID, Description, User, Timestamp, Pre-condition, Post-condition
- **User Actions:**
 1. View or search event logs.
- **System Responses:**
 - Displays system-generated event data (e.g., user registration, account update, journal approval).

5.3.10 Financial Reports Page

- **Description:** Allows Managers and Accountants to generate and export key financial statements.
- **Screen Objects:**
 - Report type dropdown (Trial Balance, Income Statement, Balance Sheet, Retained Earnings Statement)
 - Date range selector
 - “Generate Report” button
 - Output options: View, Save, Email, Print
- **User Actions:**
 1. Select report type and date range.
 2. Click “Generate Report.”
 3. Choose to view, save, email, or print the generated report.

- **System Responses:**

- The system produces the selected financial statement for the specified period.
- Displays confirmation or error messages as appropriate.

5.3.13 Dashboard Page

- **Description:** Provides consistent access to system-wide modules and user information.

- **Screen Objects:**

- Top navigation bar (persistent across all pages)
- System logo
- Logged-in user's name
- Calendar widget
- Shortcut buttons: Dashboard, Chart of Accounts, Journalizing, Ledger, Event Logs, Financial Reports
- Help button
- Financial Ratios
- Important notification (Manager only)

- **User Actions:**

1. Click navigation buttons to move between modules.
2. Use the Calendar widget for quick date reference.

- **System Responses:**

- Displays the selected module.
- Maintains user session and consistent layout across pages.

7. REQUIREMENTS MATRIX

Requirement-ID	Requirement Description	Design Component	Test Case ID
ACM-01	Admins, accountants, and managers can log in.	Login	TC-ACM-01
ACM-02	Authenticate the user and redirect to the dashboard based on the role.	Login	TC-ACM-02
ACM-03	New users can submit a registration request.	NewUser	TC-ACM-03
ACM-04	Admin must receive an email to approve a new user when they are registered.	NewUser	TC-ACM-04
ACM-05	Usernames shall follow the pattern <FirstInitial><FullLastName><MMYY>	NewUser	TC-ACM-05
ADM-01	Admin shall have access to reports of all system users.	AccountManagement	TC-ADM-01
ADM-02	Admin can suspend a user for a specific date range.	AccountManagement	TC-ADM-02
ADM-03	Admin can send an email to any user in the system.	AccountManagement	TC-ADM-03
COA-01	Admin can add, view, edit, and deactivate accounts	ChartOfAccounts	TC-COA-01
COA-02	Account must include: name, number, description, normal side, category, subcategory, initial balance, debit/credit field, running balance, date/time added, ID of creator, statement type, comment, and order.	ChartOfAccounts	TC-COA-02
COA-03	No duplicate account names or numbers	ChartOfAccounts	TC-COA-03
COA-04	Monetary values must be displayed with two decimal places and commas where appropriate	CharOfAccounts	TC-COA-04
COA-05	Accounts with a balance greater than 0 cannot be deactivated	ChartOfAccounts	TC-COA-05

COA-06	Admin can view individual account details and generate a full accounts report	ChartofAccounts	TC-COA-06
COA-07	Search and filter accounts in the chart	ChartofAccounts	TC-COA-07
COA-08	Clicking account shall navigate to that account's ledger	ChartofAccounts	TC-COA-08
COA-09	Managers and accountants cannot add, edit, or deactivate accounts.	ChartofAccounts	TC-COA-09
COA-10	Users can send emails to managers or accountant users from COA page.	ChartofAccounts	TC-COA-10
DBM-01	All login attempts, password histories, and user activities must be stored in the database	Database	TC-DBM-01
DBM-02	Event logs for each record shall be stored in the database	EventLog	TC-DBM-02
DSH-01	Landing page shall serve as the user's primary dashboard after login.	Dashboard	TC-DSH-01
DSH-02	Menu buttons on the dashboard shall correspond to the user's role permissions.	Dashboard	TC-DSH-02
DSH-03	Must dedicate a section of the landing page to display important system messages.	Dashboard	TC-DSH-03
DSH-04	Landing page's color and scheme must be visually consistent with other pages in the application.	Dashboard	TC-DSH-04
ERR-01	Error messages must be displayed in red	UserInterface	TC-ERR-01
ERR-02	Log all errors in the database table	Database	TC-ERR-02
EVL-01	For every account added, modified, or deactivated, an event log for that record shall be created and stored in DB.	EventLog	TC-EVL-01
EVL-01	Logs show before and after images, user ID, date, and time of change – all roles can view for accounts	EventLog	TC-EVL-02
FIN-01	All users must be able to view financial ratios on the landing page.	Dashboard	TC-FIN-01
FIN-02	Dashboard shall display each financial ratio	Dashboard	TC-FIN-02

	using the three-color scheme: green, yellow, and red.		
JOR-01	Managers & accountants can create new journal entries only using accounts in COA.	Journal	TC-JOR-01
JOR-02	Debits must come before credits, and multiple debit and credit entries per entry are allowed.	Journal	TC-JOR-02
JOR-03	Entries can be reset/canceled before submission, but once submitted, cannot be deleted.	Journal	TC-JOR-03
JOR-04	Journal entry must have at least one debit and one credit.	Journal	TC-JOR-04
JOR-05	Debits must equal credits.	Journal	TC-JOR-05
JOR-06	Source documents of types PDF, Word, Excel, CSV, JPG, and PNG can be attached to new entries.	Journal	TC-JOR-06
JOR-07	The manager can approve or reject entries, and rejection requires a reason in the comment field.	Journal	TC-JOR-07
JOR-08	Once entry is approved, it must be reflected in the ledger	Journal	TC-JOR-08
JOR-09	Must be able to filter entries based on status by date	Journal	TC-JOR-09
JOR-10	Must be able to view entries created by other managers and accountants.	Journal	TC-JOR-10
JOR-11	Errors must prevent the submission of invalid entries	Journal	TC-JOR-11
JOR-12	The system shall notify managers when journal entries are submitted for approval	Journal	TC-JOR-12
LED-01	Clicking the account number in a journal entry shall navigate the user to the ledger for that account.	Ledger	TC-LED-01
LED-02	Ledger entries must display the date, description, debit, credit, and balance of each transaction.	Ledger	TC-LED-02

LED-03	Each entry must include a clickable PR (Post reference) to the originating journal entry.	Ledger	TC-LED-03
LED-04	Must be able to filter entries by date, name, or amount.	Ledger	TC-LED-04
LED-05	Ledger balances must be accurate after each posting	Ledger	TC-LED-05
REP-01	Generate, view, save, email, or print Trial Balance, Income Statement, Balance Sheet, and Retained Earnings Statement for a specific date or range.	FinancialReports	TC-REP-01
SEC-01	All users are required to log in using credentials stored in DB	Login	TC-SEC-01
SEC-02	Passwords must meet complexity rules (min 8 char, include letters, numbers, special char, and must start with a letter)	NewUser	TC-SEC-02
SEC-03	Only authorized users with appropriate roles shall be able to access specific pages	Login	TC-SEC-03
SEC-04	The system shall auto-suspend the account after 3 consecutive failed login attempts.	Login	TC-SEC-04
SEC-05	System shall notify users by email 3 days before the password expires and prevent the reuse of past passwords.	AccountManagement	TC-SEC-05
SEC-06	Users must provide account email, ID, and answer security questions when resetting a password.	ForgotPassword	TC-SEC-06
UI-01	Login page shall contain fields for username, password, submit, forgot password, create new user, and company logo.	Login	TC-UI-01
UI-02	Once logged in username and profile must appear in the top right corner of all pages.	Header	TC-UI-02
UI-03	Logged in users should appear in the top left corner, company logo on all pages, consistent layout and color, tooltips, and a help button.	UserInterface	TC-UI-03
UI-04	Pop-up calendar in the left corner and	UserInterface	TC-UI-04

	navigation buttons to other pages shall appear at the top of each page.		
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